

```
10. def __init__(self):
11.     # Calculate moving averages
12.     self.sma_short = bt.indicators.SMA(self.data.close, period=self.params.short_period)
13.     self.sma_long = bt.indicators.SMA(self.data.close, period=self.params.long_period)
14.
15.     # Create a crossover indicator
16.     self.crossover = bt.indicators.CrossOver(self.sma_short, self.sma_long)
17.
18. def next(self):
19.     # If not in the market and we get a buy signal
20.     if not self.position and self.crossover > 0:
21.         self.buy()
22.
23.     # If in the market and we get a sell signal
24.     elif self.position and self.crossover < 0:
25.         self.sell()
26.
27. # Create a Cerebro engine
28. cerebro = bt.Cerebro()
29.
30. # Add the strategy
31. cerebro.addstrategy(MovingAverageCrossover)
32.
33. # Add data
34. data = bt.feeds.YahooFinanceData(dataname='AAPL', fromdate=datetime(2020, 1, 1),
todate=datetime(2023, 1, 1))
35. cerebro.adddata(data)
36.
37. # Set starting cash
38. cerebro.broker.setcash(10000.0)
39.
40. # Add analyzers
41. cerebro.addanalyzer(bt.analyzers.SharpeRatio, _name='sharpe')
42. cerebro.addanalyzer(bt.analyzers.DrawDown, _name='drawdown')
43. cerebro.addanalyzer(bt.analyzers>Returns, _name='returns')
44.
45. # Run the backtest
46. results = cerebro.run()
```